A LOCALLY FINITE TYPE MORPHISM WITH AN AMPLE INVERTIBLE SHEAF BECOMES A PROJECTIVE IMMERSION AFTER TWISTING BY A LARGE ENOUGH POWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-quasi-projective-finite-type-over-S is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 9412-9425 and proof span 9427-9515 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's aside that the first half proves the result 'for some d and n which is enough in virtually all applications' before the upgrade to all large powers; that sentence is kept verbatim.

Editorial note. 8 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The characterization of ampleness by the basis of the topology given by the affine opens defined by homogeneous forms in the section ring. Cited at proof lines 9427-9432 and again at proof lines 9483-9485. Produces the finite cover by affine opens cut out by homogeneous sections and, later, the global generation of all large powers.
- The description of sections of a power of an invertible sheaf over a distinguished affine open as quotients of homogeneous sections, and the characterization of locally finite type morphisms by finite type ring maps on affine charts. Cited at proof lines 9441-9449. Turns generators of the coordinate rings of the affine opens into homogeneous sections of a single large power.
- The construction of a morphism to projective space from a generating set of sections of an invertible sheaf, and the identification of the pullback of the twisting sheaf. Cited at proof lines 9459-9462. Produces the morphism j with the correct pullback of the twisting sheaf.
- Permanence of immersions under base change and products, the Segre embedding, and closedness of composites of immersions. Cited at proof lines 9508-9515. Shows that the morphism produced in the second half is again an immersion.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 01NF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2156–2156. Same-chapter reference used in the retained author text at source line(s) 9465. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01NE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2155–2155. Same-chapter reference used in the retained author text at source line(s) 9464. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01WD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2478–2478. Same-chapter reference used in the retained author text at source line(s) 9508, 9689, 10652. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 07RK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9870–9870. Same-chapter reference used in the retained author text at source line(s) 361, 9510, 9691, 10764. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01PW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2245–2245. Same-chapter reference used in the retained author text at source line(s) 8858, 9446. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original proposition 01Q3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2252–2252. Same-chapter reference used in the retained author text at source line(s) 8676, 8876, 9431, 9484, 9655, 10874. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02V0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3751–3751. Same-chapter reference used in the retained author text at source line(s) 180, 246, 9514, 9693. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on X. Assume that*

(1) *the invertible sheaf $\mathcal{L}$ is ample on X, and*
(2) *the morphism $X \to S$ is locally of finite type.*

Then there exists a $d_0 \geq 1$ such that for every $d \geq d_0$ there exist an $n \geq 0$ and an immersion $i : X \to \mathbf{P}^n_S$ over S such that $\mathcal{L}^{\otimes d} \cong i^*\mathcal{O}_{\mathbf{P}^n_S}(1)$.

Proof. Let $A = \Gamma_*(X, \mathcal{L}) = \bigoplus_{d \geq 0} \Gamma(X, \mathcal{L}^{\otimes d})$. By Properties, Proposition 01Q3 the set of affine opens X_a with $a \in A_+$ homogeneous forms a basis for the topology of X. Hence we can find finitely many such elements $a_0, \ldots, a_n \in A_+$ such that

(1) we have $X = \bigcup_{i=0,\ldots,n} X_{a_i}$,
(2) each X_{a_i} is affine, and
(3) each X_{a_i} maps into an affine open $V_i \subset S$.

By Lemma 01T2 we see that the ring maps $\mathcal{O}_S(V_i) \to \mathcal{O}_X(X_{a_i})$ are of finite type. Hence we can find finitely many elements $f_{ij} \in \mathcal{O}_X(X_{a_i})$, $j = 1, \ldots, n_i$ which generate $\mathcal{O}_X(X_{a_i})$ as an $\mathcal{O}_S(V_i)$-algebra. By Properties, Lemma 01PW we may write each f_{ij} as $a_{ij}/a_i^{e_{ij}}$ for some $a_{ij} \in A_+$ homogeneous. Let N be a positive integer which is a common multiple of all the degrees of the elements a_i, a_{ij}. Consider the elements

$$a_i^{N/\deg(a_i)},\ a_{ij}a_i^{(N/\deg(a_i))-e_{ij}} \in A_N.$$

By construction these generate the invertible sheaf $\mathcal{L}^{\otimes N}$ over X. Hence they give rise to a morphism

$$j : X \longrightarrow \mathbf{P}^m_S \quad \text{with } m = n + \sum n_i$$

over S, see Constructions, Lemma 01NE and Definition 01NF. Moreover, $j^*\mathcal{O}_{\mathbf{P}_S}(1) = \mathcal{L}^{\otimes N}$. We name the homogeneous coordinates $T_0, \ldots, T_n, T_{ij}$ instead of $T_0, \ldots, T_m$. For $i = 0, \ldots, n$ we have $j^{-1}(D_+(T_i)) = X_{a_i}$. Moreover, pulling back the element T_{ij}/T_i via $j^\sharp$ we get the element $f_{ij} \in \mathcal{O}_X(X_{a_i})$. Hence the morphism j restricted to X_{a_i} gives a closed immersion of X_{a_i} into the affine open $D_+(T_i) \cap \mathbf{P}^m_{V_i}$ of $\mathbf{P}^N_S$. Hence we conclude that the morphism j is an immersion. This implies the lemma holds for some d and n which is enough in virtually all applications.

This proves that for one $d_2 \geq 1$ (namely $d_2 = N$ above), some $m \geq 0$ there exists some immersion $j : X \to \mathbf{P}^m_S$ given by global sections $s'_0, \ldots, s'_m \in \Gamma(X, \mathcal{L}^{\otimes d_2})$. By Properties, Proposition 01Q3 we know there exists an integer d_1 such that $\mathcal{L}^{\otimes d}$ is globally generated for all $d \geq d_1$. Set $d_0 = d_1 + d_2$. We claim that the lemma holds with this value of d_0. Namely, given an integer $d \geq d_0$ we may choose $s''_1, \ldots, s''_t \in \Gamma(X, \mathcal{L}^{\otimes d-d_2})$ which generate $\mathcal{L}^{\otimes d-d_2}$ over X. Set $k = (m+1)t$ and denote $s_0, \ldots, s_k$ the collection of sections $s'_\alpha s''_\beta$, $\alpha = 0, \ldots, m$, $\beta = 1, \ldots, t$. These generate $\mathcal{L}^{\otimes d}$ over X and therefore define a morphism

$$i : X \longrightarrow \mathbf{P}^{k-1}_S$$

such that $i^*\mathcal{O}_{\mathbf{P}^n_S}(1) \cong \mathcal{L}^{\otimes d}$. To see that i is an immersion, observe that i is the composition

$$X \longrightarrow \mathbf{P}^m_S \times_S \mathbf{P}^{t-1}_S \longrightarrow \mathbf{P}^{k-1}_S$$

where the first morphism is (j, j') with j' given by $s''_1, \ldots, s''_t$ and the second morphism is the Segre embedding (Constructions, Lemma 01WD). Since j is an immersion, so is (j, j') (apply Lemma 07RK to $X \to \mathbf{P}^m_S \times_S \mathbf{P}^{t-1}_S \to \mathbf{P}^m_S$). Thus i is a composition of immersions and hence an immersion (Schemes, Lemma 02V0). $\square$